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Printed Page: 1 of 1

Subject Code:BP301T

Roll No:

BPHARM

(SEM. III) THEORY EXAMINATION 2025-26
PHARMACEUTICAL ORGANIC CHEMISTRY II

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

2x10 = 20

(a)	Define aromaticity.
(b)	State the conditions necessary for a compound to be aromatic.
(c)	What is sulphonation of benzene?
(d)	Define phenol and explain effect of substituents on its acidity.
(e)	Define acetyl value.
(f)	Define iodine value.
(g)	Write the structure and one use of Chloramine-T.
(h)	What are drying oils?
(i)	Define benzenoid polynuclear hydrocarbons.
(j)	State Coulson and Moffitt modification of Baeyer's strain theory.

SECTION B

2. Attempt any *two* of the following:

10 x 2 = 20

(a)	Explain nitration OR halogenation of benzene with mechanism. (Any ONE reaction)
(b)	Discuss the acidity of aromatic acids and explain the effect of electron-withdrawing and electron-donating substituents on the acidity of benzoic acid.
(c)	Explain the significance and principle involved in the determination of Acid Value, Saponification Value and Acetyl Value of fats and oils.

SECTION C

3. Attempt any *five* part of the following:

7 x 5 = 35

(a)	Effect of substituents on acidity/basicity of phenols or aromatic amines.
(b)	Explain the preparation, structure and pharmaceutical uses of BHC (Benzene Hexachloride).
(c)	Describe the preparation and important reactions of benzoic acid.
(d)	Explain preparation, properties and uses of phenols (or cresols/resorcinol).
(e)	Describe the principle, significance and determination of Reichert–Meissl (RM) value.
(f)	Explain Baeyer's strain theory, its limitations, and modifications by Coulson–Moffitt and Sachse–Mohr theory. Discuss stability of cycloalkanes.
(g)	Naphthalene OR Phenanthrene OR Anthracene (any one): structure, synthesis and medicinal importance.